# Deliverable 6 — Section 2

Integer Linear Programming Formulation of the Student–Project Assignment Problem

Group 5-6 — Team B (Assignment Engine)
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One model, three objective functions. The sets, parameters, variables and constraints (§1–§4) are shared by every strategy; only the objective (§5) changes. Tags R1–R17 refer to `docs/team_b_decisions.md`; they are references, not symbols. Every symbol below has exactly one meaning (see §7).

## 1 Index sets

| Symbol | Meaning |
|---|---|
| $S = \{s_1, \dots, s_n\}$, $\lvert S\rvert = n$ | students |
| $P = \{p_1, \dots, p_m\}$, $\lvert P\rvert = m$ | projects |
| $T$, indexed by $t$ | catalogued skills |
| $F \subseteq S \times P$ | fixed (locked) assignments |
| $E_p \subseteq T$ | mandatory ("essential") skill requirements of project $p$ |
| $T_p^{\text{soft}} \subseteq T$ | non-mandatory skill requirements of project $p$ |
| $Q_{p,t} = \{s \in S : \alpha_{s,t} \geq \beta_{p,t}\}$ | students qualified for $t \in E_p$ |
| $P^E = \{p \in P : E_p \neq \emptyset\}$ | projects with a mandatory requirement |
| $\Omega$ | strategies in order: Preference, SkillFit, Balanced |

All sets are constructed in sorted identifier order, so identical input yields an identical model (R16).

## 2 Parameters

| Symbol | Meaning |
|---|---|
| $l_p \in \mathbb{Z}_{\geq 0},\ u_p \in \mathbb{Z}_{\geq 1},\ l_p \leq u_p$ | min. and max. headcount of project $p$ |
| $\alpha_{s,t} \in \{0, 1, \dots, 5\}$ | proficiency of $s$ in skill $t$ (0 = not held) (R7) |
| $\beta_{p,t} \in \{1, \dots, 5\}$ | minimum level of skill $t$ required by $p$ (R7) |
| $K \in \mathbb{Z}_{\geq 1}$ | number of distinct alternatives requested, default 3 (R5) |

Alternatives are indexed $k = 1, \dots, K$; $j$ denotes an earlier alternative.

**Preference rank (R14).** An unranked project receives the penalty rank $m + 1$:

$$r_{s,p} = \begin{cases} \text{stated rank of } p \text{ by } s, & \text{if } s \text{ ranked } p, \\ m + 1, & \text{otherwise.} \end{cases} \tag{1}$$

**Normalised preference score.** $\phi_{s,p} \in [0, 1]$, equal to 1 for a first choice and 0 for an unranked project:

$$\phi_{s,p} = \frac{m + 1 - r_{s,p}}{m}. \tag{2}$$

**Skill fit (R4).** Computed over non-mandatory requirements only:

$$\sigma_{s,p} = \begin{cases} \dfrac{1}{|T_p^{\text{soft}}|} \displaystyle\sum_{t \in T_p^{\text{soft}}} \min\left(1, \frac{\alpha_{s,t}}{\beta_{p,t}}\right), & T_p^{\text{soft}} \neq \emptyset, \\ 1, & T_p^{\text{soft}} = \emptyset. \end{cases} \tag{3}$$

Note that $r_{s,p}$, $\phi_{s,p}$ and $\sigma_{s,p}$ are all *data*: they are computed before solving, so every objective below is linear in the decision variables.

# 3 Decision variables

| | | | |
|---|---|---|---|
| $x_{s,p} \in \{0,1\}$ | $\forall s \in S,\ p \in P$ | 1 iff student $s$ is assigned to project $p$ | (4) |
| $y_p \in \{0,1\}$ | $\forall p \in P^E$ | 1 iff project $p$ is used (R13) | (5) |
| $\theta \in \mathbb{R}_{\geq 0}$ | | worst assigned rank (BALANCED only) | (6) |

# 4 Hard constraints (all strategies)

These are never relaxed. If they cannot all hold, the run is reported as infeasible with diagnostics (R2, R10).

| | | | |
|---|---|---|---|
| $\displaystyle\sum_{p \in P} x_{s,p} = 1$ | $\forall s \in S$ | (each student exactly once, R1) | (7) |
| $l_p \leq \displaystyle\sum_{s \in S} x_{s,p} \leq u_p$ | $\forall p \in P$ | (headcount bounds, R1) | (8) |
| $x_{s,p} = 1$ | $\forall (s,p) \in F$ | (fixed assignments, R3) | (9) |
| $\displaystyle\sum_{s \in Q_{p,t}} x_{s,p} \geq y_p$ | $\forall p \in P^E,\ t \in E_p$ | (mandatory skill covered, R13) | (10) |
| $\displaystyle\sum_{s \in S} x_{s,p} \leq u_p\, y_p$ | $\forall p \in P^E$ | (non-empty $\Rightarrow$ used, R13) | (11) |

Constraint (11) forces $y_p = 1$ whenever project $p$ receives a student, which activates (10). A project with $l_p = 0$ may therefore stay empty ($y_p = 0$) without violating its mandatory requirements; if $l_p \geq 1$, (8) makes it non-empty and the requirement must be met.

**No-good cuts for distinct alternatives (R5).** Let $\Gamma^{(j)} = \{(s,p) : x_{s,p}^{(j)} = 1\}$ be the pairs of the $j$-th alternative found. Before solving for alternative $k$ ($k = 2, \ldots, K$), add for every earlier alternative

$$\sum_{(s,p) \in \Gamma^{(j)}} x_{s,p} \leq n - 1 \qquad j = 1, \ldots, k-1. \tag{12}$$

Since $|\Gamma^{(j)}| = n$, at least one student must move. If the model becomes infeasible, no further distinct alternative exists; the solutions found so far are returned and their number is reported in `diagnostics["distinct_alternatives"]`.

# 5 Objective functions (strategies)

**Order of strategies (R15).** Alternative 1 uses the strategy selected by the user; alternative $k$ uses the next strategy in $\Omega$ (cyclically) together with the cuts (12).

**Ties (R16).** Each strategy has a secondary term with a small weight; any remaining exact tie is resolved by the fixed sorted model order and a single-threaded HiGHS run.

### 5.1 Strategy 1 — Preference: minimise total rank

$$\begin{aligned} \min_{x,y} \quad & Z_{\text{pref}} = \sum_{s\in S}\sum_{p\in P} r_{s,p}\, x_{s,p} \;-\; \varepsilon_{\text{pref}} \sum_{s\in S}\sum_{p\in P} \sigma_{s,p}\, x_{s,p}, \qquad \varepsilon_{\text{pref}} = \frac{1}{n+1} \\ \text{s.t.} \quad & (7)\text{–}(11),\ \text{cuts } (12) \end{aligned} \tag{13}$$

The rank sum is an integer and the skill term is bounded by $n\,\varepsilon_{\text{pref}} < 1$, so the secondary term only breaks ties and can never worsen the rank sum. Minimising the rank sum is equivalent to minimising the mean rank $\bar{r}$.

### 5.2 Strategy 2 — SkillFit: maximise total skill fit

$$\begin{aligned} \max_{x,y} \quad & Z_{\text{skill}} = \sum_{s\in S}\sum_{p\in P} \sigma_{s,p}\, x_{s,p} \;+\; \varepsilon_{\text{skill}} \sum_{s\in S}\sum_{p\in P} \phi_{s,p}\, x_{s,p}, \qquad \varepsilon_{\text{skill}} = \frac{10^{-3}}{n} \\ \text{s.t.} \quad & (7)\text{–}(11),\ \text{cuts } (12) \end{aligned} \tag{14}$$

The preference term is at most $10^{-3}$. Because skill-fit totals are not integers, this is a practical (not exact) lexicographic guarantee; we state this as a limitation.

### 5.3 Strategy 3 — Balanced: protect the worst-off, then balance

Solved in two stages, so no weight between fairness and efficiency has to be tuned.

**Stage 1 (egalitarian / min–max rank):**

$$\begin{aligned} \theta^* = \min_{x,y,\theta} \quad & \theta \\ \text{s.t.} \quad & \theta \geq \sum_{p\in P} r_{s,p}\, x_{s,p} \qquad \forall s \in S \\ & (7)\text{–}(11),\ \text{cuts } (12) \end{aligned} \tag{15}$$

**Stage 2 (balance among min–max optimal solutions):**

$$\begin{aligned} \max_{x,y} \quad & Z_{\text{bal}} = \sum_{s\in S}\sum_{p\in P} \big(\lambda\,\phi_{s,p} + (1-\lambda)\,\sigma_{s,p}\big)\, x_{s,p}, \qquad \lambda = 0.5 \\ \text{s.t.} \quad & \sum_{p\in P} r_{s,p}\, x_{s,p} \leq \theta^* \qquad \forall s \in S \\ & (7)\text{–}(11),\ \text{cuts } (12) \end{aligned} \tag{16}$$

Both $\phi_{s,p}$ and $\sigma_{s,p}$ lie in $[0,1]$, so $\lambda = 0.5$ weighs them equally. Stage 1 guarantees that no student receives a worse rank than strictly necessary.

## 6 Complete model at a glance (Strategy 1 shown)

$$
\begin{aligned}
\min \quad & \sum_{s\in S}\sum_{p\in P}\Big(r_{s,p} - \tfrac{1}{n+1}\,\sigma_{s,p}\Big)\,x_{s,p} && \\
\text{s.t.} \quad & \sum_{p\in P} x_{s,p} = 1 && \forall s \in S \\
& l_p \le \sum_{s\in S} x_{s,p} \le u_p && \forall p \in P \\
& x_{s,p} = 1 && \forall (s,p) \in F \\
& \sum_{s\in Q_{p,t}} x_{s,p} \ge y_p && \forall p \in P^E,\ t \in E_p \\
& \sum_{s\in S} x_{s,p} \le u_p\, y_p && \forall p \in P^E \\
& \sum_{(s,p)\in\Gamma^{(j)}} x_{s,p} \le n-1 && j = 1,\dots,k-1 \\
& x_{s,p} \in \{0,1\}, \quad y_p \in \{0,1\} &&
\end{aligned}
$$

**Size.** $nm + |P^E|$ binary variables; $n + 2m + |F| + \sum_{p\in P^E}(|E_p|+1) + (k-1)$ constraints (plus $n$ for each BALANCED stage).

## 7 Notation summary

| Symbol | Meaning |
|---|---|
| $S, s, n$ | students, a student, number of students |
| $P, p, m$ | projects, a project, number of projects |
| $T, t$ | skills, a skill |
| $T_p^{\text{soft}}$, $E_p$ | non-mandatory / mandatory requirements of $p$ |
| $Q_{p,t}$, $P^E$ | qualified students; projects with a mandatory requirement |
| $F$ | fixed assignments |
| $\Omega$ | ordered list of strategies |
| $K, k, j$ | number of alternatives, an alternative, an earlier alternative |
| $\Gamma^{(j)}$ | student–project pairs of alternative $j$ |
| $l_p, u_p$ | minimum / maximum headcount |
| $\alpha_{s,t}$, $\beta_{p,t}$ | proficiency; required level |
| $r_{s,p}$, $\phi_{s,p}$, $\sigma_{s,p}$ | preference rank; preference score; skill fit |
| $x_{s,p}$, $y_p$, $\theta$ | assignment; project used; worst rank |
| $Z_{\text{pref}}, Z_{\text{skill}}, Z_{\text{bal}}$ | objective functions |
| $\varepsilon_{\text{pref}}, \varepsilon_{\text{skill}}, \lambda$ | tie-break weights; balance weight |